

ARES: An Automated Evaluation Framework for Retrieval-Augmented Generation Systems

Jon Saad-Falcon

Stanford University^{*}

jonsaadfalcon@stanford.edu

Omar Khattab

Stanford University

okhattab@stanford.edu

Christopher Potts

Stanford University

cgpotts@stanford.edu

Matei Zaharia

Databricks and UC Berkeley

matei@databricks.com

Abstract

Evaluating retrieval-augmented generation (RAG) systems traditionally relies on hand annotations for input queries, passages to retrieve, and responses to generate. We introduce ARES, an *Automated RAG Evaluation System*, for evaluating RAG systems along the dimensions of context relevance, answer faithfulness, and answer relevance. By creating its own synthetic training data, ARES finetunes lightweight LM judges to assess the quality of individual RAG components. To mitigate potential prediction errors, ARES utilizes a small set of human-annotated datapoints for prediction-powered inference (PPI). Across eight different knowledge-intensive tasks in KILT, SuperGLUE, and AIS, ARES accurately evaluates RAG systems while using only a few hundred human annotations during evaluation. Furthermore, ARES judges remain effective across domain shifts, proving accurate even after changing the type of queries and/or documents used in the evaluated RAG systems. We make our code and datasets publicly available on [Github](#).

The best design for a RAG system is not necessarily universal across data domains, corpus sizes, and cost/latency budgets. To tune their own RAG systems, practitioners traditionally need hand annotations for test questions, passages to retrieve (to assess the retriever), and responses to generate, labeled specifically for their target domain. Alternatively, they may evaluate different approaches in production by collecting human preferences that compare the candidate systems. Unfortunately, both of these strategies demand high expertise and impose considerable annotation costs.

Model-based evaluation is an inexpensive strategy to test generative output quality (Zheng et al., 2023). For instance, the open-source RAGAS framework (James and Es, 2023) prompts an LM for evaluating the *relevance* of retrieved information and the *faithfulness* and *accuracy* of generated responses. Unfortunately, such strategies currently rely for evaluation on a fixed set of heuristically hand-written prompts, offering little adaptability to various evaluation contexts and no guarantees about quality.

To evaluate RAG systems rapidly and accurately, we propose ARES, the **Automated RAG Evaluation System**. ARES is the first automated RAG evaluation system to generate tailored LLM judges for each component of a RAG pipeline, leading to substantial boosts in evaluation precision and accuracy compared to existing approaches like RAGAS. Furthermore, unlike existing RAG evaluation systems, ARES provides confidence intervals for its scoring by leveraging prediction-powered inference (PPI; Angelopoulos et al. 2023). Given a corpus of documents and a RAG system, ARES reports three evaluation scores: context relevance (is the retrieved information pertinent to the test question), answer faithfulness (is the response generated by the language model properly grounded in the retrieved context), and answer relevance (is the response also relevant to the question). A good

1 Introduction

Retrieval-augmented generation (RAG) has become a prominent approach for building user-facing NLP applications, such as systems for question answering (QA), fact-checking, and customer support (Petroni et al., 2021; Wang et al., 2019). Typically, a RAG system consists of a retriever and a downstream language model (LM). Given a user question, the retriever finds relevant passages from a corpus and the LM uses these passages to generate a response. This formulation admits a multitude of choices: what retrieval model to use, how to divide the documents into retrieval chunks, and how to prompt or finetune the LM to use the retrieved information, to name only a few of the simplest design decisions.

^{*}Project started during research internship at Databricks

RAG system finds relevant contexts and generates answers that are both faithful and relevant.

Many existing RAG evaluation frameworks require substantial human annotations for scoring. ARES significantly improves data efficiency during evaluation by only requiring three inputs: an in-domain passage set, a human preference validation set of approximately 150 annotated datapoints or more, and few-shot examples of in-domain queries and answers (e.g. five examples or more), which are used for prompting LLMs in synthetic data generation.

Given the corpus of in-domain passages, ARES proceeds in three stages. First, it leverages an LM to construct a synthetic dataset of question-answer pairs, derived from the passages in the corpus. Second, it defines three separate judge models to perform three classification tasks (context relevance, answer faithfulness, and answer relevance). These judges are lightweight models fine-tuned against a contrastive learning objective. Third, ARES scores the different RAG systems being assessed using prediction-powered inference (PPI; Angelopoulos et al. 2023) to improve model-based evaluation accuracy and provide statistical confidence intervals for RAG scoring. PPI utilizes a small set of human annotated datapoints for computing its confidence intervals; we designate this annotated set as our *human preference validation set*, which is composed of approximately 150 annotated datapoints or more that designate both positive and negative examples for context relevance, answer faithfulness, and answer relevance.

We conduct extensive empirical evaluations, demonstrating that ARES accurately scores RAG systems across the six knowledge-intensive datasets in KILT and SuperGLUE, beating existing automated evaluation approaches like RAGAS by 59.3 and 14.4 percentage points on average across context relevance and answer relevance evaluation accuracy, respectively. Additionally, ARES accurately calculates answer hallucination occurrences in the AIS attribution dataset (Rashkin et al., 2022), predicting within 2.5 percentage points of the ground truth average for answer hallucinations. Compared to annotation-based evaluation methods, ARES is substantially more accurate and efficient, requiring 78% less annotations than the baseline approach. We also find that ARES consistently distinguishes competitive RAG systems that are only a few points apart in ground-truth metrics. This precision enables ARES to guide the develop-

ment and comparison of competitive approaches and configurations.

We make the ARES code and datasets publicly available on Github.

2 Related Work

RAG (Guu et al., 2020; Lewis et al., 2020; Khat-tab et al., 2021; Izacard et al., 2022)) is now a common strategy for bolstering LLMs by combining them with retrieval systems. Through retrieval, RAG helps LM systems gather domain-specific knowledge, ground generations in factual information (Shuster et al., 2021; Huo et al., 2023), and offer a degree of transparency or interpretability via citing sources (Mialon et al., 2023).

Multiple LLM-based evaluation techniques have emerged for gauging LLM systems. This is essential for rapid deployment in new settings, where it is difficult to build a traditional benchmark dataset from scratch. Early attempts at this use LLMs out of the box, as in MT-Bench and Chatbot Arena (Zheng et al., 2023). AutoCalibrate (Liu et al., 2023b) seeks to align an LLM-judge with human preferences, leveraging a self-refinement prompt to iteratively improve the LLM judge. However, AutoCalibrate does not offer any statistical guarantees for the accuracy of its predictions. Other work has used LLM prompting to evaluate system quality across natural language generation tasks, such as translation, summarization, and dialogue (Kocmi and Federmann, 2023; Fu et al., 2023; Liu et al., 2023a; Wang et al., 2023).

In the context of knowledge-intensive NLP tasks, LLMs have been explored for assessing attribution and factuality in LLMs (Min et al., 2023; Gekhman et al., 2023; Yue et al., 2023). New guidelines like LongEval (Krishna et al., 2023) and datasets like Hagrid and ALCE (Kamalloo et al., 2023; Gao et al., 2023) provide resources for analyzing knowledge-intensive LLM pipelines.

The two most-closely related projects to ARES are EXAM (Sander and Dietz, 2021) and RAGAS (James and Es, 2023). To evaluate RAG systems, the EXAM metric estimates how many exam questions a reader (simulated as a QA system) can answer correctly based on the generated response. This requires a set of queries with several associated sub-questions each, which adds a burden that ARES does not bring. RAGAS is based on a handful of heuristic hand-written prompts. These offer little adaptability to new RAG evaluation set-

tings (e.g., new corpora) and, as we show in our evaluation, substantially underperform ARES.

3 ARES

ARES proceeds in three stages (Figure 1). There are three required inputs: an in-domain passage set, a human preference validation set of approximately 150 annotated datapoints (or more), and few-shot examples of in-domain queries and answers (five or more examples), which are used for prompting LLMs in synthetic data generation. With our inputs prepared, we begin by generating synthetic queries (and their answers) from the passages in the target corpus. We then use these query–passage–answer triples to train LLM judges. Subsequently, we apply these judges to any RAG system, scoring a sample of its in-domain query–document–answer triples, and use prediction-powered inference (PPI) with our human preference validation set to estimate a confidence interval for the quality of each RAG system.

3.1 LLM Generation of Synthetic Dataset

We generate synthetic queries and answers from the corpus passages using generative LLMs. The generated data represent both positive and negative examples of query–passage–answer triples (e.g., relevant/irrelevant passages and correct/incorrect answers). For generation, the LLM uses our input set of few-shot examples with in-domain passages mapped to in-domain queries and answers; the model then generates a synthetic question and answer from a given in-domain passage, allowing us to create both positive and negative training examples. We include example prompts for generating synthetic queries and answers in A.6.

For creating our synthetic data, we primarily use on FLAN-T5 XXL (discussed in subsection 4.1). ARES works well with this model (see section 5) but our system can ultimately use another high-quality model for generating synthetic queries and answers. We then filter out low-quality queries by testing if a given query can retrieve its original passage as the top result using its retriever. This filtering approach has been used in previous work to isolate high-quality synthetic queries (Dai et al., 2022; Saad-Falcon et al., 2023).

To generate negatives for fine-tuning our LLM judges, we rely on two novel strategies, generating the same number of negatives with each strategy:

1. **Weak Negative Generation:** For context rel-

evance negatives, we randomly sample in-domain passages unrelated to a given synthetic query. For answer faithfulness and answer relevance negatives, we randomly sample synthetically-generated answers from other passages, which were created using FLAN-T5 XXL.

2. **Strong Negative Generation:** For context relevance negatives, we randomly sample in-domain passages from the same document as the gold passage. For datasets in which multiple passages are not available for the same document, we use BM25 to retrieve the top-10 passages similar to the passage and sample from them for our context relevance strong negatives. For answer faithfulness and answer relevance negatives, we prompt FLAN-T5 XXL (subsection 4.1) to generate a contradictory answer using the few-shot prompt in subsection A.5.

In total, the number of negatives generated equals the number of positives generated for evaluating context relevance and answer relevance.

3.2 Preparing LLM Judges

To prepare our RAG evaluation judges, we use our synthetic dataset to fine-tune DeBERTa-v3-Large judges (discussed in subsection 4.1) to evaluate three different capabilities (Chen et al., 2023; James and Es, 2023):

1. **Context Relevance:** Is the passage returned relevant for answering the given query?
2. **Answer Faithfulness:** Is the answer generated faithful to the retrieved passage, or does it contain hallucinated or extrapolated statements beyond the passage?
3. **Answer Relevance:** Is the answer generated relevant given the query and retrieved passage?

For each metric, a separate LLM with a binary classifier head is fine-tuned to classify positive and negative examples. For each concatenated query–document–answer, a single LLM judge must classify the triple as positive or negative for that judge’s metric. To fine-tune these judges, we use our human preference validation set to evaluate model improvement after each epoch, stopping when we have three epochs with no improvement in loss (see subsection A.1 for more information).

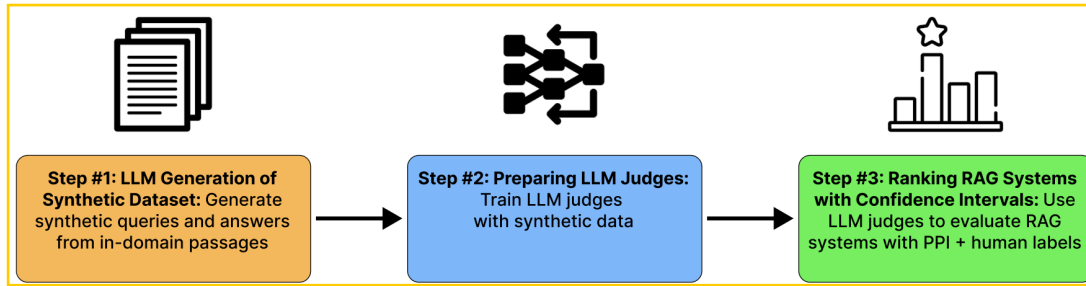


Figure 1: **Overview of ARES:** As inputs, the ARES pipeline requires an in-domain passage set, a human preference validation set of 150 annotated datapoints or more, and few-shot examples of in-domain queries and answers (five or more), which are used for prompting LLMs in synthetic data generation. To prepare our LLM judges for evaluation, we first generate synthetic queries and answers from the corpus passages. Using our generated training triples and a contrastive learning framework, we fine-tune an LLM to classify query–passage–answer triples in three different criteria: context relevance, answer faithfulness, and answer relevance. Finally, we use the LLM judges to score RAG systems and generate confidence bounds for the ranking using PPI and the human preference validation set.

3.3 Ranking RAG Systems with Confidence Intervals

Once we have prepared our LLM judges, we need to use them to score and rank the competing RAG systems. To do this, ARES samples the in-domain query-document-answer triples produced by each RAG approach, and the judges label each triple, predicting their context relevance, answer faithfulness, and answer relevance. By averaging the individual predicted labels for each in-domain triple, we calculate the RAG system performance across each of the three metrics.

In principle, we could simply report these average scores as quality metrics for each RAG system. However, these scores reflect entirely unlabeled data with predictions from a synthetically-trained LLM judge, and hence they may not be entirely accurate. As an extreme alternative, we could use just the small human preference validation set discussed previously for evaluation, reporting the extent to which each RAG system agrees with (or deviates from) the human annotations. However, an annotation-based evaluation approach would require labeling substantially more generative outputs from each RAG systems separately, which can be costly both in terms of time and financing.

To combine the benefits of both, and hence boost the precision of the evaluation, ARES uses *prediction-powered inference* (PPI; Angelopoulos et al. 2023) to predict the system scores. PPI is a recent statistical method that provides tighter confidence intervals on a small set of annotated datapoints (i.e., our validation set) by leveraging predictions on a much larger set of non-annotated datapoints. PPI can leverage both the labeled dat-

apoints and the ARES judge predictions on the non-annotated datapoints to construct confidence intervals for our RAG system’s performance.

To do this, PPI uses the LLM judges on the human preference validation set to learn a *rectifier function* for constructing a confidence set of the ML model’s performance, using each ML prediction in the larger non-annotated dataset. The confidence set can then be used to create a tighter confidence interval for the performance of the evaluated RAG system (e.g. its context relevance, answer faithfulness, or answer relevance accuracy individually) compared to simply using annotated outputs from the evaluated RAG system. By bolstering the human preference validation set with the much larger set of datapoints with ML predictions, PPI can develop reliable confidence intervals for ML model performance that beat previous classical inference approaches.

The PPI rectifier function allows us to estimate the errors of the LLM judge and generate confidence bounds for the success and failure rates of the RAG system, estimating context relevance, answer faithfulness, and answer relevance performance. Additionally, PPI allows us to estimate confidence intervals with a selected level of probability; for our experiments, we use a standard 95% alpha (probability) for our confidence interval.

With the accuracy confidence interval for each component of the RAG, we find the midpoint of each confidence interval and use the midpoints to rank the RAG systems. With our ranking, we can compare different RAG systems, as well as different configurations of the same RAG system, to find the best-performing approach for a given domain.

4 Experiments

4.1 Models

For our fine-tuned judges, ARES relies on generating cheap but quality synthetic queries and answers using LLMs. For generating our synthetic datasets, we use FLAN-T5 XXL (Chung et al., 2022). We selected DeBERTa-v3-Large (He et al., 2021) for our fine-tuned LLM judge. Our fine-tuned LLM judges allow us to rank RAG systems without relying on external APIs, solely using few-shot prompts and deployable LLMs on commercial GPUs.

For our in-context learning baseline, we use OpenAI’s *gpt-3.5-turbo-16k*, version 10/23, (Brown et al., 2020) in a zero/few-shot setting. For similarity search over in-domain passages, we use FAISS IndexFlatL2 for indexing (Johnson et al., 2019) and OpenAI’s *text-embedding-ada-002* for generating embeddings. We use similarity search over in-domain passages to filter our synthetic queries that cannot retrieve the passage from which they were generated. We use version 0.0.18 of RAGAS in our experiments (James and Es, 2023).

4.2 Datasets

Our core experimental goal is to provide a rich picture of where ARES can be applied effectively. To test across multiple types of queries, documents, and answers, we selected all the datasets from the widely-used KILT and SuperGLUE benchmarks for which RAG is appropriate.

From KILT (Petroni et al., 2021), we use Natural Questions (NQ), HotpotQA, FEVER, and Wizards of Wikipedia (WoW) (Kwiatkowski et al., 2019; Yang et al., 2018; Akhtar et al., 2023; Dinan et al., 2018). Each dataset uses Wikipedia passages but the queries and answers offer a range of applications. Both NQ and HotpotQA feature direct questions and expect short answers, but NQ uses single passages for reasoning while HotpotQA requires multiple passages for reasoning. Furthermore, FEVER focuses on fact-verification, determining if a passage supports or refutes a given statement, and expects an output of “SUPPORTS” or “REFUTES”. WoW seeks to evaluate dialogue agents by mapping user dialogue to relevant Wikipedia passages before a chatbot generates a paragraph-length chat response incorporating passage knowledge.

From SuperGLUE (Wang et al., 2019), we use MultiRC and ReCoRD (Khashabi et al., 2018; Zhang et al., 2018). MultiRC focuses on direct questions for seven different domains (News,

Wikipedia articles, articles on society/law/justice, articles on history/anthropology, elementary school science textbooks, 9/11 reports, and fiction). ReCoRD focuses on determining the placeholder entity in a statement, focusing on news articles from CNN and the Daily Mail. For MultiRC and ReCoRD, we create open-domain versions of their tasks. For MultiRC, we perform retrieval over its seven sets of domain passages. For ReCoRD, we perform retrieval over its news article passages.

The efficacy of ARES relies on its ability to rank different RAG systems while only using a human preference validation set and domain-targeted LLM judges. To test the limits of ARES, we need to simulate the existence of many RAG systems that are separated by small accuracy margins on our evaluation metrics. For this, we create systems using artificial query-passage-answer triples, in which we empirically know the positive and negative examples of the mock RAG system. We generate these mock splits of the given datasets by selecting (1) The positive and negative query-passage matches for context relevance, and (2) the positive and negative query-passage-answer matches for answer relevance. We include positive and negative examples from our evaluation sets in Table 7.

For our positive triples, we can simply use the KILT and SuperGLUE examples without any alteration. For gathering negative query-passage pairs and query-passage-answer triples, we randomly sample passages and answers from either: the same Wikipedia document or an entirely random Wikipedia document. This sampling allows us to artificially create mock RAG systems for testing ARES. By sampling both related and unrelated documents/answers, we hope to better gauge the efficacy of ARES in judging RAG outputs.

We do not evaluate answer faithfulness for KILT and SuperGLUE datasets since we do not have human-annotated hallucinated answers to use for evaluation. However, we do test the ARES framework on real attribution datasets in Section 5.2.

Using the validation subsets for each KILT and SuperGLUE dataset, we create nine different dataset splits, ranging from 70% success rate to 90% success rate for each of the evaluated RAG criteria; each dataset is separated by 2.5% accuracy points (e.g. 70.0%, 72.5%, 75.0%, ..., 90.0%). Each split also represents a different mock RAG system. Since we know the success percentages of each dataset split, we know the appropriate ranking of each mock RAG system. This allows us to

test ARES success at both scoring and ranking the mock RAG systems appropriately across the three evaluation criteria.

4.3 Metrics

To calculate the correlation between the correct ranking and the ARES ranking, we use the Kendall rank correlation coefficient or Kendall's τ :

$$\tau = \frac{(\# \text{ of concordant pairs}) - (\# \text{ of discordant pairs})}{\# \text{ of pairs total}}$$

Concordant pairs are defined as two ordinal values in the ranking where the earlier value in the sequence is lower than the later value in the sequence. Discordant pairs are defined as two ordinal values in the ranking where the earlier value in the sequence is greater than or equal to the later value in the sequence. A Kendall's τ greater than 0.9 is considered successful but it ranges from 0.0 to 1.0.

In development, researchers and engineers will be comparing different RAG configurations through individual pairwise comparisons of model choices, retriever selection, and document preprocessing. We want to make sure that ARES has satisfactory accuracy in pairwise comparisons across a variety of performance gaps between RAG systems. Kendall's τ is explicitly designed for measuring the accuracy of such pairwise comparisons, calculating the correlation between a perfectly accurate pairwise ranking and an experimental pairwise ranking. Thus, it is a popular and widespread metric used in information retrieval, allowing developers to evaluate ranking systems empirically. Therefore, we believe Kendall's tau and prediction accuracy provide meaningful metrics for testing the efficacy of ARES as a RAG evaluation system.

5 Results & Analysis

5.1 ARES Ranking

Table 1 summarizes our main evaluation of ARES (with DeBERTa-v3-Large as the pretrained basis for the judges). We compare against RAGAS (version 0.0.18) and a baseline few-shot prompted GPT-3.5 judge (*gpt-3.5-turbo-16k*). For the few-shot GPT-3.5 judge, we provide few-shot examples for guiding predictions; the prompts are included in Appendices A.2, A.3, and A.4. For both ARES and the GPT-3.5 judge baseline, we augment the LLM with PPI, using a 300-datapoint human preference validation set to rectify the ML predictions and produce confidence intervals.

Across almost all settings across the datasets from KILT and SuperGLUE, ARES provides a more accurate ranking of RAG systems than RAGAS. ARES averages a Kendall's τ 0.065 higher for context relevance and 0.132 higher for answer relevance than RAGAS. Additionally, the LLM-judge is substantially more accurate than RAGAS at predicting context relevance and answer relevance of a query-passage-answer triple. For context relevance, ARES with a fine-tuned LLM-judge is 59.9 percentage points higher than RAGAS while for answer relevance, our system is 14.4 percentage points higher than RAGAS. Overall, ARES provides a more accurate system for automatically evaluating RAG configurations than RAGAS by leveraging domain-adaptive techniques for prompting and training as well as utilizing PPI to bolster model predictions.

As an additional comparison, we also include the Kendall's τ for RAG ranking with the ARES LLM judge without PPI; for all datasets tested, PPI improved the ranking prediction accuracy of the fine-tuned LLM judge. Furthermore, we included a sampled annotations configuration, in which we sampled 150-datapoints from each mock RAG system, totalling 1,350 annotations. Even with all these annotations, the Kendall's τ for ARES is 0.08 higher on average, across both context and answer relevance, compared to sampled annotations, despite using 78% less annotations. In sum, ARES proves significantly more data-efficient with human annotations while being more accurate at scoring than standard sampled annotation methods.

Compared to the GPT-3.5 judge, ARES provides a more accurate ranking of the RAG systems than the GPT-3.5 judge, averaging a Kendall's tau 0.06 higher over both context relevance and answer relevance. Between the judge configurations, the fine-tuned LLM judge of ARES can more precisely distinguish between RAG systems and guide configuration decisions surrounding document splitting, retriever selection, and generative LLM choice. However, while the fine-tuned LLM judge had a higher Kendall's tau on average, the GPT-3.5 judge is more readily deployable and does not require any additional fine-tuning. The GPT-3.5 judge does come with its own querying costs, which can vary based on the date of querying as well as the total tokens used in evaluation.

We also wanted to better understand the importance of human annotations for ARES. To this end, we conducted two sets of experiments. First, we

	ARES Ranking of Pseudo RAG Systems											
	NQ		HotpotQA		WoW		FEVER		MultiRC		ReCoRD	
	C.R.	A.R.	C.R.	A.R.	C.R.	A.R.	C.R.	A.R.	C.R.	A.R.	C.R.	A.R.
Kendall’s Tau for Sampled Annotations	0.83	0.89	0.78	0.78	0.78	0.83	0.89	0.89	0.83	0.83	0.72	0.94
Kendall’s Tau for RAGAS	0.89	0.89	0.94	0.89	0.94	0.94	0.72	0.61	0.83	0.94	0.89	0.44
Kendall’s Tau for GPT-3.5 Judge	0.89	0.94	0.67	0.94	0.94	0.89	0.78	0.78	0.83	0.89	0.83	0.94
Kendall’s Tau for ARES LLM Judge	0.89	1.0	0.89	0.94	0.94	1.0	0.83	0.72	0.94	0.83	0.78	0.83
Kendall’s Tau for ARES	0.94	1.0	0.94	0.94	1.0	1.0	0.89	0.78	0.94	0.89	0.83	0.89
RAGAS Accuracy	31.4%	71.2%	17.2%	76.0%	36.4%	77.8%	23.7%	69.2%	16.1%	75.0%	15.0%	72.8%
GPT-3.5 Judge Accuracy	73.8%	95.5%	75.3%	71.6%	84.3%	85.2%	60.4%	59.6%	72.4%	60.3%	81.0%	65.8%
ARES Accuracy	79.3%	97.2%	92.3%	81.3%	85.7%	96.1%	88.4%	78.5%	85.8%	82.7%	67.8%	92.3%

Table 1: **ARES Ranking with Fine-tuned LLM Judges vs. Sampled Annotations, RAGAS and GPT-3.5 Judge:** For scoring context relevance and answer relevance (C.R. and A.R. in the table, respectively), we compare ARES with our fine-tuned LLM judges against sampled annotations benchmark, RAGAS, and a few-shot GPT-3.5 judge. For our sampled annotations, we gather 150 annotated datapoints from each mock RAG system and use those labels to score the system. RAGAS also uses GPT-3.5 as its judge but it uses few-shot prompts that are not targeted for each evaluation domain. Overall, we found that ARES ranked RAG systems more accurately than RAGAS and GPT-3.5 across all the explored datasets. The Kendall’s tau for ARES was 0.065 higher on average for scoring context relevance and 0.132 higher on average for scoring answer relevance than RAGAS. Additionally, we include the Kendall’s taus for the ARES LLM Judge without PPI and found that PPI further boosted the ranking accuracy of the judge across the board. We selected GPT-3.5 instead of GPT-4 due to the lower financial costs required to run. For PPI in both ARES and the GPT-3.5 judge, we used 300 human annotations for our human preference validation set. The prompts used for the GPT-3.5 judges are included in Sections A.2, A.3, and A.4.

used ARES with human annotation sets ranging in size from 25 to 400 and found that 150 is the minimum number required (Table 3). Second, we explored whether GPT-4 generations could replace human annotations entirely, finding that GPT-4 is less good than humans in this role, though the idea arguably has promise (Table 4).

5.2 ARES Performance on AIS

	WoW	CNN / DM
ARES Split Prediction	0.478	0.835
Correct Positive/Negative Split	0.458	0.859
ARES Judge Accuracy	62.5%	84.0%
Evaluation Set Size	707	510
Human Preference Data Size	200	200

Table 2: ARES Results on the AIS benchmark

To evaluate whether ARES can effectively gauge answer faithfulness in real RAG systems, we tested ARES on the AIS attribution benchmark (Rashkin et al., 2022). In AIS, we selected the Wizards of Wikipedia (WoW) and CNN/DM datasets; the

other benchmark datasets involve either table reasoning (ToTTo) or focus on passage summarization (QRECC) so we excluded them. In WoW and CNN/DM, each evaluation example includes a query, a retrieved passage, and a generated answer (which is either faithful or non-attributed to the retrieved passage).

Table 2 summarizes our AIS results. We found that ARES can effectively score the AIS datasets, getting within 2.5 accuracy points of the correct scores. Furthermore, for scoring each system, we only use 200 annotated datapoints for our human preference validation set. Our results on AIS demonstrate the ability of ARES to reliably distinguish faithful and hallucinated answers in real-world RAG systems.

5.3 ARES Ranking of Existing RAG Systems

We also wanted to evaluate whether ARES can score and rank existing RAG systems across both context relevance and answer relevance. For evaluation, we selected the NQ, WoW, and FEVER datasets from KILT. We consider the answer gen-

erations to be correct if they contained the KILT answer in their output. For our RAG systems, we selected three different retrievers (BM25, OpenAI Ada embeddings with cosine similarity search, and ColBERTv2 (Santhanam et al., 2022)) and three different generative LLMs (MPT-7b-Instruct (Team, 2023), GPT-3.5-Turbo, and GPT-4). Additionally, we include the Facebook RAG model (Lewis et al., 2020), which uses a DPR retriever (Karpukhin et al., 2020) and BART sequence-to-sequence model (Lewis et al., 2019). During retrieval, each RAG system only retrieves one passage to assist generation.

In Table 5, we found that ARES can reliably score and rank RAG systems in real-world applications, averaging a Kendall’s tau of 0.91 for context relevance and 0.97 for answer relevance. Compared to RAGAS, ARES is 0.16 higher for context relevance and 0.15 higher for answer relevance, on average. ARES also provided accurate confidence bounds for its predictions, capturing the ground truth average outcomes for context relevance and answer relevance more than 95% of the time; on average, the PPI confidence intervals were 7.4 points wide for context relevance and 6.1 points wide for answer relevance (see Figure 2 and Figure 3 for ARES vs. RAGAS). Among the models tested, the best performing retriever was ColBERTv2 while the best performing generative LLM was GPT-4.

5.4 Strengths and Limits of Cross-Domain

Applications

The generalizability of the LLM judge used in ARES is critical for deploying our framework in specialized domains, particularly domains where in-domain queries, documents, and answers are difficult to gather. Therefore, we wanted to test how the LLM judges used in ARES would be affected by three domain shifts: change in *query type* from training to test (e.g. NQ to FEVER), change in *document type* from training to test (e.g. NQ to MultiRC), and change in both *query and document type* (e.g. NQ to ReCoRD).

In Table 6, we found that the fine-tuned LLM judges used in ARES proved successful in cross-domain applications. Across all settings, we found that LLM judges in ARES had strong generalizability, even when only using 300 datapoints in our human preference validation set for PPI. Furthermore, we found that even when the LLM judge’s accuracy suffered in cross-domain applications, PPI helped mitigate the loss in accuracy and still allow

ARES to be successful. Additional examples for PPI also continued to boost cross-domain ARES performance in subsequent tests.

While LLM judges in ARES were successful in cross-domain applications for KILT and SuperGLUE, LLM judges are unable to generalize when making more drastic shifts in domain, such as: switching languages (e.g. English to Spanish, German, and other languages), switching from text to code (e.g. questions + passages to coding functions + documentation), and switching from retrieving text to extraction of entities, webpages, or citations.

To test cross-lingual transfer, we used the XGLUE datasets (Liang et al., 2020); a LLM judge fine-tuned on NQ achieved a Kendall’s tau of 0.33 over both context relevance and answer relevance scoring for XGLUE. To test text-to-code, we used CodeSearchNet (Husain et al., 2019); an LLM judge fine-tuned on NQ achieved a Kendall’s tau of 0.28 over both context relevance and answer relevance scoring for CodeSearchNet. To test extraction task generalizability, we used T-Rex from KILT (Elsahar et al., 2018; Petroni et al., 2021); an LLM judge fine-tuned on NQ achieved a Kendall’s tau of 0.38 over both context relevance and answer relevance scoring for T-Rex. Each cross-domain shift requires in-domain passages and few-shot query examples for reconfiguring ARES judges.

6 Conclusion

In this work, we present ARES, a novel automated evaluation framework for retrieval-augmented generation (RAG). ARES offers a novel training pipeline for fine-tuning lightweight LLM judges on synthetically generated queries and answers. ARES can evaluate each component of a RAG system separately to help improve system understanding and create targeted solutions, and it requires only minimal human annotations. For the eight different datasets in KILT, SuperGLUE, and AIS requiring RAG-based solutions, we found that ARES can accurately score and rank RAG systems based on context relevance, answer faithfulness, and answer relevance scores, beating the existing RAGAS automated evaluation framework.

ARES is a flexible framework, and there may be variants of it that are even more powerful than the ones we explored here. Avenues to explore include GPT-4 as a replacement for human labeling (Table 4), more robust techniques for the synthetic datasets used in fine-tuning LLM judges, utilizing

logits in LLM judge prediction to improve PPI confidence intervals, and testing more sophisticated LLMs as fine-tuned judges for ARES.

7 Limitations

ARES relies on a small set of annotations in the human preference validation set (roughly 150-300 datapoints but more is better). These annotations often require an annotator familiar with the RAG system’s domain application. While these annotations can be easy to generate for general-domain applications, more specialized domains, such as law, medicine, and finance, may require annotators with specialized expertise.

The LLMs used in ARES benefit substantially from GPU-based hardware with substantial storage. In ARES, DeBERTa-v3-Large (304M) and FLAN-T5-XXL (11.3B) required GPUs with about 32GB of memory to run, taking several hours for fine-tuning and generation, respectively. While commercial GPUs are widely available, they are not easily accessible to all NLP researchers and practitioners due to their costs.

Additionally, all of the datasets used in our evaluation of ARES are in English, a well-resourced language with abundant annotations. Future work should explore how ARES can be employed in other languages by utilizing different LLMs for the ARES judge and the synthetic data generation. This can help us better understand the strengths and weaknesses of the current ARES framework.

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A Appendix

A.1 Fine-tuning Configuration for LLM

Judges

For our loss function used in LLM judge training, we selected cross-entropy loss using Adam

(Kingma and Ba, 2017). For our classification head, we use a single linear classification layer and apply a 0.1 dropout to the input, which is the final hidden state of the [CLS] token. For our learning schedule, we use linear warmup and linear decay (Howard and Ruder, 2018) with a 5e-6 learning rate and a 32 training batch size across all experimental configurations.

A.2 GPT Prompting for Context Relevance

Scoring

For the NQ, HotpotQA, MultiRC, and ReCoRD datasets, we use 8 few-shot examples with the following prompt to score context relevance:

- Given the following question and document, you must analyze the provided document and determine whether it is sufficient for answering the question. In your evaluation, you should consider the content of the document and how it relates to the provided question. Output your final verdict by strictly following this format: "[[Yes]]" if the document is sufficient and "[[No]]" if the document provided is not sufficient. Do not provide any additional explanation for your decision.

Question: *<few-shot example here>*

Document: *<few-shot example here>*

For FEVER, we use the following prompt to score context relevance:

- You are an expert fact-checking agent. Given the following statement and document, you must analyze the provided document and determine whether it is sufficient for determining the statement’s factuality. In your evaluation, you should consider the content of the document and how it relates to the provided statement’s factuality. Output your final verdict by strictly following this format: "[[Yes]]" if the document is sufficient and "[[No]]" if the document is not sufficient. Do not provide any additional explanation for your decision.

Statement: *<few-shot example here>*

Document: *<few-shot example here>*

For WoW, we use the following prompt to score context relevance:

- You are an expert dialogue agent. Given the following dialogue and document, you must

analyze the provided document and determine whether it is relevant for responding to the dialogue. In your evaluation, you should consider the content of the document and how it relates to the provided dialogue. Output your final verdict by strictly following this format: "[[Yes]]" if the document is relevant and "[[No]]" if the document provided is not relevant. Do not provide any additional explanation for your decision.

Dialogue: <few-shot example here>

Document: <few-shot example here>

A.3 GPT Prompting for Answer Faithfulness

Scoring

For the NQ, HotpotQA, MultiRC, and ReCoRD datasets, we use 8 few-shot examples with the following prompt to score answer faithfulness:

- Given the following question, document, and answer, you must analyze the provided answer and determine whether it is faithful to the contents of the document. The answer must not offer new information beyond the context provided in the document. The answer also must not contradict information provided in the document. Output your final verdict by strictly following this format: "[[Yes]]" if the answer is faithful to the document and "[[No]]" if the answer is not faithful to the document. Do not provide any additional explanation for your decision.

Question: <few-shot example here>

Document: <few-shot example here>

Answer: <few-shot example here>

For FEVER, we change the word "question" in the prompt to "statement". For WoW, we change the word "question" in the prompt to "dialogue".

A.4 GPT Prompting for Answer Relevance

Scoring

For the NQ, HotpotQA, MultiRC, and ReCoRD datasets, we use 8 few-shot examples with the following prompt to score answer relevance:

- Given the following question, document, and answer, you must analyze the provided answer and document before determining whether the answer is relevant for the provided question. In your evaluation, you should consider

whether the answer addresses all aspects of the question and provides only correct information from the document for answering the question. Output your final verdict by strictly following this format: "[[Yes]]" if the answer is relevant for the given question and "[[No]]" if the answer is not relevant for the given question. Do not provide any additional explanation for your decision.

Question: <few-shot example here>

Document: <few-shot example here>

Answer: <few-shot example here>

For FEVER, we change the word "question" in the prompt to "statement". For WoW, we change the word "question" in the prompt to "dialogue".

A.5 Prompting for Generation of Synthetic Queries and Answers

To generate synthetic queries and answers using FLAN-T5, we use the following prompt and provide 5 few-shot examples:

- Example N

Question: <few-shot example here>

Document: <few-shot example here>

Answer: <few-shot example here>

We use the same prompting structure for generating incorrect or contradictory answers; we simply swap out the few-shot examples to be incorrect or contradictory instead.

A.6 Synthetic Query and Answer Generation

For generating our synthetic questions, we use the following prompt for FLAN-T5 XXL:

- Example #1

Document: <few-shot example here>

Query: <few-shot example here>

Example #2

Document: <few-shot example here>

Query: <few-shot example here>

Example #3

Document: <few-shot example here>

Query: <few-shot example here>

Example #4

Document: <in-domain passage>

Query:

For generating our synthetic answers, we use the following prompt for FLAN-T5 XXL:

- Example #1

Query: *<few-shot example here>*

Document: *<few-shot example here>*

Answer: *<few-shot example here>*

Example #2

Query: *<few-shot example here>*

Document: *<few-shot example here>*

Answer: *<few-shot example here>*

Example #3

Query: *<few-shot example here>*

Document: *<few-shot example here>*

Answer: *<few-shot example here>*

Example #4

Query: *<synthetic query here>*

Document: *<in-domain passage here>*

Answer:

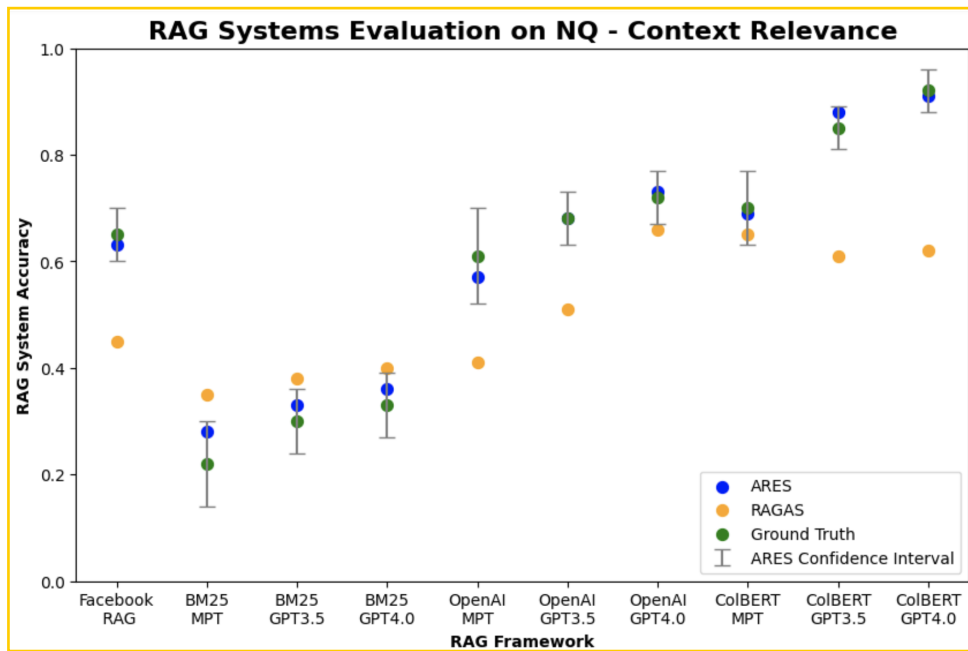


Figure 2: RAG Systems Evaluation on NQ - Context Relevance

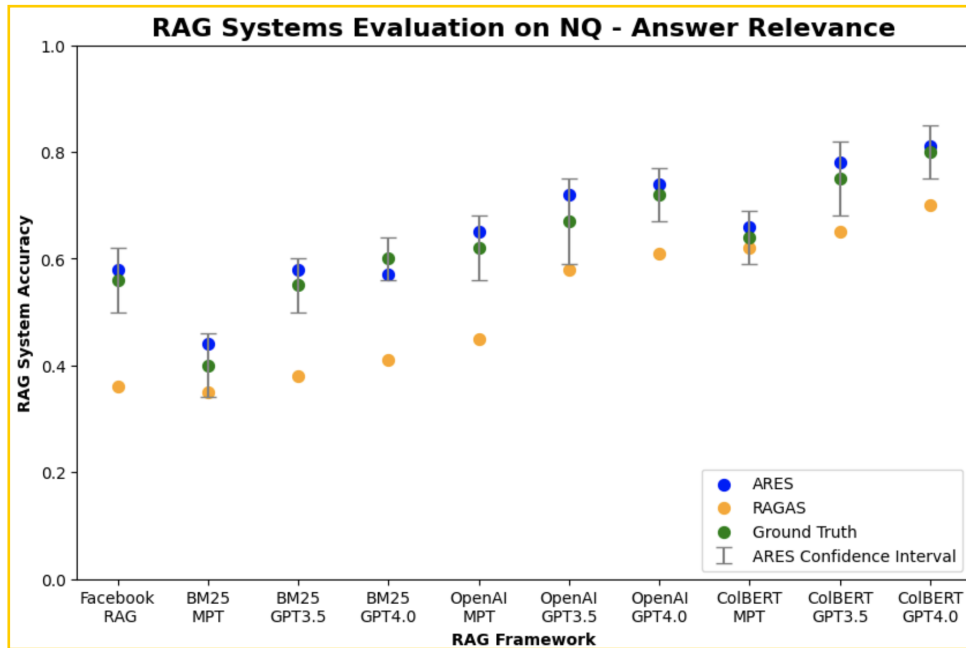


Figure 3: RAG Systems Evaluation on NQ - Answer Relevance

PPI Labeled Count	Kendall's Tau by Dataset					
	NQ		MultiRC		ReCoRD	
	C.R.	A.R.	C.R.	A.R.	C.R.	A.R.
400	1.0	1.0	0.89	0.94	0.89	0.94
300	0.89	1.0	0.94	0.89	0.83	0.89
200	0.83	1.0	0.83	0.94	0.83	0.83
150	0.72	1.0	0.83	0.89	0.72	0.83
100	0.44	1.0	0.67	0.67	0.67	0.83
50	0.44	0.94	0.61	0.44	0.56	0.67
25	0.44	0.89	0.56	0.44	0.44	0.56

Table 3: **Analysis of PPI Labeled Count vs. ARES Efficacy by Kendall's Tau:** The Kendall's tau values represent the correlation between the correct ranking and the ARES ranking of the pseudo RAG systems. We use the same experimental set-up as described in subsection 4.2. We find that below about 100-150 datapoints in the human preference validation set, ARES cannot meaningfully distinguish between the alternate RAG systems based on their accuracies in context relevance and answer relevance (C.R. and A.R., respectively).

	ARES Ranking of Pseudo RAG Systems using GPT-4 Labels					
	NQ		ReCoRD		MultiRC	
	Context Relevance	Answer Relevance	Context Relevance	Answer Relevance	Context Relevance	Answer Relevance
Kendall's Tau	0.78	1.0	0.78	0.72	0.89	0.78
Kendall's Tau of Human Labeled Approach	0.94	1.0	0.83	0.89	0.94	0.89
Average PPI Range	9.2%	6.8%	8.2%	9.0%	7.7%	8.3%
Accuracy on RAG Evaluation Sets	79.3%	96.7%	88.4%	78.3%	85.8%	82.5%

Table 4: **GPT-4 Labels vs. Human Labels:** We wanted to explore the practicality of using GPT-4 generated labels instead of human annotations for our human preference validation set in ARES. In the experiments, we generated 500 GPT-4 labels as replacements for human labeling using few-shot prompts (see Sections A.2, A.3, and A.4). While GPT-4 generated labels decreased Kendall's tau in most settings by 0.05 to 0.30, the ability to cheaply produce GPT-4 generated labels significantly reduces the cost of annotation, cutting it from hundreds of annotations to less than ten for few-shot prompts. Additionally, the efficacy of PPI continues improving as we generate more GPT-4 generated labels. In the table, we define PPI range as the number of percentage points from the lower number to the upper number of the PPI confidence bounding. Additionally, we use the fine-tuned LLM judge (DeBERTa-v3-Large) for evaluation.

	ARES Ranking of Real RAG Systems					
	NQ		WoW		FEVER	
	C.R.	A.R.	C.R.	A.R.	C.R.	A.R.
Kendall’s Tau for Sampled Annotations	0.73	0.78	0.73	0.73	0.73	0.82
Kendall’s Tau for RAGAS	0.82	0.82	0.73	0.82	0.73	0.87
Kendall’s Tau for GPT-3.5 Judge	0.82	0.87	0.82	0.82	0.64	0.87
Kendall’s Tau for ARES LLM Judge	0.91	0.96	0.91	1.0	0.73	0.87
Kendall’s Tau for ARES	1.0	0.96	0.91	1.0	0.82	1.0
RAGAS Accuracy	35.9%	68.2%	44.4%	80.1%	21.4%	75.9%
GPT-3.5 Accuracy	80.5%	91.2%	81.2%	83.5%	61.3%	54.5%
ARES Accuracy	85.6%	93.3%	84.5%	88.2%	70.4%	84.0%

Table 5: **ARES Ranking on Real-World RAG Systems:** For scoring context relevance and answer relevance (C.R. and A.R. in the table, respectively), we compare ARES with our fine-tuned LLM judges against sampled annotations benchmark, RAGAS, and a few-shot GPT-3.5 judge. For our sampled annotations, we gather 150 annotated datapoints from each mock RAG system and use those labels to score the system. RAGAS also uses GPT-3.5 as its judge but it uses few-shot prompts that are not targeted for each evaluation domain. Overall, we found that ARES ranked RAG systems more accurately than RAGAS and GPT-3.5 across all the explored datasets. Additionally, we include the Kendall’s taus for the ARES LLM Judge without PPI and found that PPI further boosted the ranking accuracy of the judge across the board. We selected GPT-3.5 instead of GPT-4 due to the lower financial costs required to run. For PPI in both ARES and the GPT-3.5 judge, we used 300 human annotations for our human preference validation set. The prompts used for the GPT-3.5 judges are included in Sections A.2, A.3, and A.4.

	ARES Cross-Domain Ranking of Pseudo RAG Systems											
	NQ to FEVER		FEVER to NQ		NQ to MultiRC		MultiRC to NQ		NQ to ReCoRD		ReCoRD to NQ	
	C.R.	A.R.	C.R.	A.R.	C.R.	A.R.	C.R.	A.R.	C.R.	A.R.	C.R.	A.R.
Kendall’s Tau	0.89	0.89	1.0	0.83	0.94	0.89	1.0	0.89	0.78	0.89	0.89	0.94
Kendall’s Tau of In-Domain LLM Judge	0.89	0.78	0.94	1.0	0.94	0.89	0.94	1.0	0.83	0.89	0.94	1.0
Average PPI Range	8.7%	7.2%	6.5%	11.5%	10.2%	11.3%	11.9%	11.5%	10.5%	10.1%	9.7%	6.2%
Accuracy on RAG Evaluation Sets	92.4%	28.4%	85.7%	22.6%	81.5%	92.1%	87.6%	80.2%	29.1%	81.2%	80.1%	92.1%

Table 6: **Cross-Domain Usage of Fine-tuned LLM Judges:** We tested the cross-domain application of the fine-tuned LLM judge in the ARES framework. We found that for both context relevance and answer relevance (C.R. and A.R. in the table, respectively), fine-tuned LLM judges showed strong generalizability across domains when changing query type (e.g. NQ and FEVER), document type (e.g. NQ and MultiRC), or both (e.g. NQ and ReCoRD). For PPI, we used 300 labeled examples for our human preference validation set but also found that additional examples further improved the performance of ARES. Furthermore, we found that even in scenarios where the fine-tuned LLM judge’s accuracy significantly dropped out-of-domain (e.g. answer relevance for NQ to FEVER), PPI mitigated the decrease in judge performance. In the table, we define PPI range as the number of percentage points from the lower bound to the upper bound of the PPI confidence interval.

Query	Passage	Answer	Context Relevance	Answer Relevance
How can a ball that is not moving possess energy of position?	Mechanical energy is a combination of the energy of motion or position. This type of energy describes objects that are moving or could move. A moving ball can have energy from motion. An arrow can also have the energy of motion. Both are types of mechanical energy.	The ball holds mechanical energy	1	1
Who has a Jimmy Stewart-like quality of quiet trust?	One look at Fred Rooney, and you just know he's the good guy. A trace of childish innocence in his face gives the lanky Bethlehem lawyer a Jimmy Stewart-like quality of quiet trust. In black jeans and button-down shirt, he's a kind of folk hero in the south Bethlehem melting pot where he's crafted a law practice catering to working-class families - mostly Latino - in the shadow of the hulking remnants of Bethlehem Steel.	Fred Rooney	1	1
Before he murder the doctor and Ralph Smith, where did the stepfather reside?	Surviving being shot and stabbed at the end of the previous film , the stepfather has been institutionalized in Puget Sound, Washington since , spending his time building model houses in the workshop. Assigned a new doctor named Joseph Danvers the stepfather begins confiding in him to gain his trust , ultimately murdering the doctor during a session by stabbing him in the neck with a blade smuggled out of the workshop . After killing Danvers the stepfather beats a suspicious guard named Ralph Smith to death with his own nightstick with only two strikes and takes his uniform , successfully sneaking out of the sanitarium . Checking into a hotel after robbing and murdering a traveling salesman the stepfather alters his appearance , takes the name Doctor Gene F. Clifford from the newspaper obituaries and travels to Palm Meadows , Los Angeles after seeing an ad for it on an episode of Dream House .	Los Angeles	1	0
What was the name of the 2006 film about Pushkin's death, and who portrayed Pushkin?	After arriving in New York City, Einstein was taken to various places and events, including Chinatown, a lunch with the editors of the New York Times, and a performance of Carmen at the Metropolitan Opera, where he was cheered by the audience on his arrival. During the days following, he was given the keys to the city by Mayor Jimmy Walker and met the president of Columbia University, who described Einstein as "The ruling monarch of the mind." Harry Emerson Fosdick, pastor at New York's Riverside Church, gave Einstein a tour of the church and showed him a full-size statue that the church made of Einstein, standing at the entrance.	Vasily Szaitshev portrayed Pushkin in the film Pushkin Returns	0	0

Table 7: Positive and Negatives Evaluation Examples